

Agarwal, Dahleh, et al. (2023) extend this approach to more general missingness patterns by finding block missingness patterns embedded within more complicated missingness patterns. However, as discussed in Section 2.3, even embedded block missingness patterns are often insufficient to identify all cohort outcome means in our settings of interest, and when multiple such patterns exist, it is unclear how to efficiently combine the information from each. Our approach addresses both challenges by combining information across the many possible ways to express cohort-specific factor estimates with respect to a common basis.

Our assumption that the O^3 graph algorithm converges to a single super cohort resembles those in the literature on fixed-effect models of bipartite match outcomes under strict exogeneity (Abowd, Creecy, and Kramarz, 2002; Bonhomme et al., 2019; Hull, 2018; Jochmans and Weidner, 2019; Kline, 2024); see Bonhomme (2020) for a review. In Appendix F.2, we show that our O^3 Algorithm convergence requirement generalizes theirs to factor models with arbitrary rank.

Our consistency and asymptotic normality results relate to the literature on perturbation theory for eigenspaces of random matrices (Z. Bai and Silverstein, 2010). Standard bounds like the Davis–Kahan theorem (Yu, Wang, and Samworth, 2015) are too coarse to characterize asymptotic distributions. Instead, we derive an exact, first-order eigenspace projection expansion using Kato’s integral (Kato, 1949). Several other papers have also applied perturbation-theoretic results to show that eigenspaces of random matrices concentrate, e.g. Babii, Ghysels, and Pan (2025); Onatski (2009); Simons (2023), as well as Oliveira (2010) and Lei (2020), which consider random matrices with (approximately) independent entries.

2 Setup and Intuition

2.1 Short Panel Settings

In our settings of interest, researchers observe a large, i.i.d. sample of N units, each with T outcomes, where Y_{it}^* denotes unit i ’s outcome t . To describe which outcomes are observed, we group units into C cohorts, where C_i denotes unit i ’s cohort and $p_c := \mathbb{P}(C_i = c)$ denotes the share of units in cohort c . For cohort c , let $\mathcal{T}_c \subseteq [T] := \{1, \dots, T\}$ denote the indices of their observed outcomes and $T_c := |\mathcal{T}_c|$ denote how many are observed. We allow cohort membership C_i to be correlated with outcomes Y_{it}^* in a structured way made precise below. To distinguish observed from missing outcomes, we define $Y_{it} = Y_{it}^*$ if $t \in \mathcal{T}_{C_i}$ and $Y_{it} = ?$ otherwise.⁵ Our asymptotic theory in Section 4.2 focuses on short panels where T remains bounded as N grows. Section 6.2 discusses the more general “sparse panel” setting where T_c remain bounded but C (and hence T) can grow with N .

This setup encapsulates several causal panel data settings. In event study settings like Figure 1a’s, Y_{it}^* represents unit i ’s period- t control potential outcome—their outcome in period t had they not yet received treatment by period t —and a unit belongs to cohort c if they were first treated

⁵In what follows, we define $? \cdot 0 = 0$.

at time c . Since control potential outcomes are only observed prior to treatment, post-treatment outcomes are missing: $\mathcal{T}_c = \{t \in [T] : t < c\}$.

In bipartite matching settings, Y_{it}^* denotes “row-type” unit i ’s outcome when matched with “column-type” unit t , and row-type units belong to the same cohort if matched with the same subset of column-type units. Figure 1b illustrates such a setting, which serves as our main running example and the focus of Section 5’s empirical application. There, row-type units i are workers, t indexes pairs of a two-year period and a group of similar firms in the same Veneto province of Italy, and Y_{it}^* denotes the log of worker i ’s average weekly wage at firms in group t during t ’s two-year period. Since workers are employed by only a subset of firms, counterfactual wages at firms in groups they never worked for are missing.

Given these unbalanced panel data, researchers often seek to estimate aggregations of cohort outcome means $\mu_{ct} := \mathbb{E}[Y_{it}^* \mid C_i = c]$. We call μ_{ct} a counterfactual outcome mean since outcome t may not be observed for cohort c . Our approach enables estimation of aggregation parameters $\theta := h(\mu, \eta)$, where μ collects all cohort outcome means μ_{ct} and η is a vector of nuisance parameters estimable from observed outcomes, covariates, and additional data.

In event study settings, a common θ is a vector of average treatment effects at different relative time horizons post-treatment. As introduced in Callaway and Sant’Anna (2021) and L. Sun and Abraham (2021) and reviewed in Example D.1 of Appendix D.1, these estimands can be expressed as weighted averages of differences between each cohort’s counterfactual post-treatment control potential outcome means μ_{ct} and their observed post-treatment treated potential outcome means (part of η), with weights depending on relative cohort sizes p_c (also part of η).⁶

In the context of a bipartite matching setting with Medicare patients as row-type units and geographic areas as column-type units, Finkelstein et al. (2016) propose estimands θ that decompose differences in average observed outcomes between two column-type units into components attributable to column-type unit effects versus row-type unit selection. Below, we define a nonparametric generalization of their attribution parameters that we estimate in Section 5 using data from Figure 1b’s setting.⁷

Example 1 (Match Outcome Attribution). Analogous to Finkelstein et al. (2016), the difference in average observed match outcomes for row-type units matched to two column-type units t_1 and t_2 can

⁶See De Chaisemartin and d’Haultfoeuille (2020) for treatment effect definitions when treatment is not absorbing, and De Chaisemartin and d’Haultfoeuille (2024) when a unit’s entire treatment history may affect future outcomes.

⁷The continuation of Example 1 in Appendix D.1 shows how our generalization reduces to Finkelstein et al. (2016)’s parameters under a TWFE model (2). Hull (2018) and Kline (2024) study similar parameters under TWFE-like assumptions. We leave decompositions based on higher-order moments like variances (Abowd et al., 1999) to future work.

be decomposed additively as follows:

$$\begin{aligned}
& \mathbb{E}[Y_{it_1} \mid t_1 \in \mathcal{T}_{C_i}] - \mathbb{E}[Y_{it_2} \mid t_2 \in \mathcal{T}_{C_i}] \\
&= \underbrace{\mathbb{E}[Y_{it_1}^*] - \mathbb{E}[Y_{it_2}^*]}_{\text{Average effect on outcomes of column-type unit differences under at-random matches}} \\
&+ \underbrace{(\mathbb{E}[Y_{it_1}^* \mid t_1 \in \mathcal{T}_{C_i}] - \mathbb{E}[Y_{it_1}^*]) - (\mathbb{E}[Y_{it_2}^* \mid t_2 \in \mathcal{T}_{C_i}] - \mathbb{E}[Y_{it_2}^*])}_{\text{Average effect on outcomes of differential row-type unit selection into matches}}.
\end{aligned}$$

Following [Finkelstein et al. \(2016\)](#), we define $\theta_{\text{col}, t_1, t_2}$ as the “share” of the difference in average observed outcomes between t_1 and t_2 attributable to how these column-type units affect outcomes and $\theta_{\text{row}, t_1, t_2}$ as the share attributable to differential row-type unit selection:

$$\theta_{\text{col}, t_1, t_2} := \frac{\mathbb{E}[Y_{it_1}^*] - \mathbb{E}[Y_{it_2}^*]}{\mathbb{E}[Y_{it_1}^* \mid t_1 \in \mathcal{T}_{C_i}] - \mathbb{E}[Y_{it_2}^* \mid t_2 \in \mathcal{T}_{C_i}]} =: 1 - \theta_{\text{row}, t_1, t_2}.$$

We put “share” in quotes because, as in [Finkelstein et al. \(2016\)](#), θ_{col} and θ_{row} sum to one but need not lie between zero and one.

To express $\theta_{\text{col}, t_1, t_2}$ and $\theta_{\text{row}, t_1, t_2}$ as functions of μ and η , note that both depend on unconditional outcome means $\mathbb{E}[Y_{it}^*]$ and observed outcome means $\mathbb{E}[Y_{it}^* \mid t \in \mathcal{T}_{C_i}]$, which are both different cohort size-weighted averages of cohort outcome means:

$$\mathbb{E}[Y_{it}^*] = \sum_{c=1}^C p_c \mu_{ct}, \quad \mathbb{E}[Y_{it}^* \mid t \in \mathcal{T}_{C_i}] = \frac{\mathbb{E}[\mathbb{1}\{t \in \mathcal{T}_{C_i}\} Y_{it}^*]}{\mathbb{P}(t \in \mathcal{T}_{C_i})} = \sum_{c=1}^C \frac{\mathbb{1}\{t \in \mathcal{T}_c\} p_c}{\sum_{c'=1}^C \mathbb{1}\{t \in \mathcal{T}_{c'}\} p_{c'}} \mu_{ct}.$$

Thus, the nuisance parameters η in this example are simply the shares of units in each cohort p_c .

2.2 Factor Model of Outcomes

To estimate averages of counterfactual outcomes not observed for some cohorts, we impose additional structure on how units’ observed and latent characteristics affect their outcomes. Specifically, we assume outcomes Y_{it}^* follow a rank- r linear factor model: outcomes are determined by the inner product of an outcome-specific factor vector $\gamma_t \in \mathbb{R}^r$ and a unit-specific loading vector $\lambda_i \in \mathbb{R}^r$, plus an optional outcome fixed effect γ_{t0} , an optional term involving an always-observed covariate vector $X_{it} \in \mathbb{R}^q$, and a mean-zero error ε_{it} :

$$Y_{it}^* = \gamma_t' \lambda_i + \gamma_{t0} + X_{it}' \beta + \varepsilon_{it}, \quad \mathbb{E}[\varepsilon_{it} \mid \lambda_i, X_i, C_i] = 0, \tag{1}$$

where X_i is a $T \times q$ matrix with t -th row X_{it} . Mean-independence of the residuals ε_{it} from cohort membership C_i conditional on λ_i and X_i implies that only the coordinates of λ_i can serve as unobserved confounders affecting both cohort membership and outcomes.⁸ We assume the rank $r < T$

⁸We treat factor vectors γ_t as fixed, conditioning inference on them. If γ_t and γ_{t0} were random, the residual mean condition in (1) would read $\mathbb{E}[\varepsilon_{it} \mid \lambda_i, X_i, C_i, \{(\gamma_t, \gamma_{t0}) : t \in [T]\}] = 0$. As discussed in Section 4.3, mean independence of ε_{it} from λ_i is sometimes stronger than necessary, but it makes clear the role that factor loadings λ_i

is known throughout, though Section 6.1 discusses principled ways to empirically evaluate different values of r .

For intuition, consider Figure 1b’s worker-firm setting. There, λ_i ’s coordinates could measure worker i ’s skills (e.g., cognitive and interpersonal), and γ_t ’s coordinates could measure how firms in t ’s group value each skill, as in Lazear (2009). The outcome fixed effect γ_{0t} , if present, could capture firm-specific wage determinants independent of worker characteristics, and X_{it} could include (i, t) -specific wage determinants like worker i ’s age at the beginning of t ’s period or the interaction of i ’s education with the industry of t ’s firm group. To the extent workers sort into firms that value their skills, λ_i and X_{it} determine both which firms they work at ($\mathcal{T}_{C_i}$) and the wages they earn (Y_{it}^*) (Lazear, 2009).

Before continuing, it is worth contrasting the factor model (1) with the canonical Two-Way Fixed Effects (TWFE) model,⁹ which instead specifies outcomes as the sum of unidimensional unit and outcome fixed effects $\lambda_i, \gamma_{t0} \in \mathbb{R}$ plus covariate and error terms $X'_{it}\beta$ and ε_{it} interpreted as in (1):

$$Y_{it}^* = \lambda_i + \gamma_{t0} + X'_{it}\beta + \varepsilon_{it}, \quad \mathbb{E}[\varepsilon_{it} \mid \lambda_i, X_i, C_i] = 0. \quad (2)$$

As discussed in the event study and bipartite matching literatures (Arkhangelsky et al., 2021; Athey et al., 2021; Bonhomme et al., 2019), when outcome fixed effects are included, the factor model (1) generalizes the TWFE model (2) in two key ways.¹⁰ First, even with $r = 1$, the factor model allows the latent characteristic λ_i to affect different outcomes differently via interaction with γ_t , whereas the TWFE model requires λ_i to affect all outcomes identically. In Figure 1b’s setting, where λ_i is often interpreted as worker i ’s human capital, the TWFE model’s additive separability rules out complementarity between worker ability and firm productivity in determining wages.¹¹ Second, while the TWFE model requires confounders to affect outcomes through a unidimensional index λ_i , the factor model with $r \geq 2$ accommodates multidimensional confounding effects, e.g. different firms valuing workers’ skill bundles differently in Figure 1b’s setting.

2.3 Intuition for Factor Model Identification

Our identification approach proceeds in two steps: first, we recover factor vectors for all outcomes in a common basis; second, we use these to extrapolate from each cohort’s observed outcome means to their counterfactual outcome means. The second step is conceptually straightforward: given factor vectors $\tilde{\gamma}_t = Q\gamma_t$ rotated by some common $r \times r$ basis matrix Q , regressing unit i ’s observed outcomes Y_{it} for $t \in \mathcal{T}_c$ on the corresponding $\tilde{\gamma}_t$ yields an unbiased estimate $\tilde{\lambda}_i$ of $(Q')^{-1}\lambda_i$ (Agarwal, Dahleh, et

play as unobserved confounders in our settings of interest. Section 4.3 also discusses additional restrictions on the residuals’ covariance structure (e.g. independence across outcomes and homoskedasticity) and covariate exogeneity conditions enabling estimation of cohort-specific factor vectors.

⁹Variants of the TWFE model can be derived from and imply “parallel trends” assumptions in both event study and bipartite matching settings (Borusyak et al., 2024; Ghanem et al., 2022; Hull, 2018; Kline, 2024).

¹⁰Even without outcome fixed effects, the TWFE model (2) is a special case of (1) with $r = 2$, $\lambda_{i1} = 1$, and $\gamma_{t2} = 1$.

¹¹Even with log-wages, the TWFE model rules out super-multiplicative complementarity between worker ability and firm productivity.

2.4 The Aggregated Projection Matrix

To describe our automatic solution to the factor misalignment problem, we introduce additional notation. Let Γ be the $T \times r$ matrix with t -th row equalling γ_t , and let $\tilde{\Gamma}_c$ be the $T \times r$ matrix whose t -th row equals $\tilde{\gamma}_{ct} = Q_c \gamma_t$ if $t \in \mathcal{T}_c$ and zero otherwise. For a subset of outcomes $\mathcal{T} \subseteq [T]$, let $E_{\mathcal{T}}$ be the $T \times T$ diagonal matrix with t -th entry equalling one if $t \in \mathcal{T}$ and zero otherwise, and define $E_c := E_{\mathcal{T}_c}$. For any matrix M , let $\Pi(M) := M(M'M)^+M'$ denote the projection onto M 's column space, where M^+ denotes the Moore-Penrose pseudo-inverse of M .

Two observations underpin our solution. First, since the column space of $\tilde{\Gamma}_c = E_c \Gamma Q'_c$ does not depend on Q_c , we have $\Pi(\tilde{\Gamma}_c) = \Pi(E_c \Gamma)$ —the projection onto the column space of the matrix $E_c \Gamma$ whose non-zero rows are the true factor vectors for cohort c 's observed outcomes.¹⁶ Thus, $\Pi(\tilde{\Gamma}_c)$ provides a basis-free representation of information about cohort c 's factor vectors, uncontaminated by the cohort-specific Q_c .

Our second observation is that Γ 's column space lies in the null space of $E_c - \Pi(E_c \Gamma)$ for each cohort c .¹⁷ This fact motivates defining a particular matrix we call an *Aggregated Projection Matrix* (APM):¹⁸

$$A := \frac{1}{C} \sum_{c=1}^C [E_c - \Pi(E_c \Gamma)]. \quad (3)$$

Since A is the mean of $E_c - \Pi(E_c \Gamma) = E_c - \Pi(\tilde{\Gamma}_c)$ across cohorts, Γ 's column space lies in A 's null space. In Section 4.1, we show that, with sufficient overlap between cohorts' observed outcomes, A 's null space is *exactly* Γ 's column space. Thus, the rows of any basis matrix for A 's null space provide valid factor vectors for the downstream regression used to impute counterfactual outcomes.

3 Estimation and Inference Procedure

Having explained the conceptual underpinnings of our approach, we now describe our estimation and inference procedure, which has four steps and is summarized in Algorithm 1. A software package implementing the procedure is available at <https://github.com/brad-ross/apm>.

In step one, we use one of the methods discussed in Section 4.3—each imposing different additional assumptions—to construct estimates $\hat{\gamma}_{ct}$ of the cohort-specific factor vectors $\tilde{\gamma}_{ct}$ for each observed outcome $t \in \mathcal{T}_c$ and cohort c . We stack these estimates into a $T \times r$ matrix $\hat{\Gamma}_c$ whose t -th row is $\hat{\gamma}_{ct}$ if $t \in \mathcal{T}_c$ and $\mathbf{0}_r$ otherwise. The cohort-specific factor estimators we consider use only data from each cohort and thus can be computed efficiently in parallel.¹⁹

In step two, we use the cohort-specific estimated factor matrices $\{\hat{\Gamma}_c : c \in [C]\}$ to construct an

¹⁶By definition, $\Pi(\tilde{\Gamma}_c) = E_c \Gamma Q'_c (Q_c \Gamma' E_c \Gamma Q_c)^+ Q_c \Gamma' E_c = E_c \Gamma (\Gamma' E'_c \cdot E_c \Gamma)^+ \Gamma' E_c = \Pi(E_c \Gamma)$.

¹⁷ $\Pi(E_c \Gamma) = E_c \Pi(E_c \Gamma) = \Pi(E_c \Gamma) E_c$ and $E_c^2 = E_c$, so $(E_c - \Pi(E_c \Gamma)) \Gamma = E_c (I - \Pi(E_c \Gamma)) E_c \Gamma = \mathbf{0}_{T \times r}$.

¹⁸Despite its name, the APM is not itself a projection matrix.

¹⁹In Section 6.3, we discuss sharing data across cohort-specific factor estimators, leaving details to future work.

Algorithm 1: Estimation

Data: $\{(C_i, Y_i, X_i)\}_{i=1}^N$

- 1 **for** $c \in \{1, \dots, C\}$ **do**
 - Compute cohort-specific factor vector estimates $\hat{\gamma}_{ct}$ for $t \in \mathcal{T}_c$ (see Section 4.3)
 - Construct a $T \times r$ matrix $\hat{\Gamma}_c$ with row t equalling $\hat{\gamma}_{ct}$ if $t \in \mathcal{T}_c$, and $\mathbf{0}_r$ otherwise
 - end**
 - 2 Construct estimated APM $\hat{A}$ as in (4)
 - 3 Compute $T \times r$ matrix $\hat{\Gamma}$ of eigenvectors for $\hat{A}$'s r smallest eigenvalues
 - 4 Estimate the outcome imputation ingredients $(\hat{\gamma}_0, \hat{\beta}, \{\hat{\lambda}_i\}_{i=1}^N)$ via the regression (5)
 - 5 **for** $c \in \{1, \dots, C\}$ **do**
 - Compute outcome mean estimate vector $\hat{\mu}_c$ for cohort c as in (6)
 - end**
 - 6 Compute estimate $\hat{\eta}$ of nuisance parameters η necessary for estimating θ
 - 7 Compute plug-in target parameter estimate $\hat{\theta} = h(\hat{\mu}, \hat{\eta})$, where $\hat{\mu} = (\hat{\mu}'_1, \dots, \hat{\mu}'_C)'$
-

estimated APM:

$$\hat{A} := \frac{1}{C} \sum_{c=1}^C [E_c - \Pi(\hat{\Gamma}_c)]. \quad (4)$$

Let $\hat{\Gamma}$ be the $T \times r$ matrix whose columns are eigenvectors corresponding to the r smallest eigenvalues of $\hat{A}$.²⁰ Row t of $\hat{\Gamma}$, denoted $\hat{\gamma}_t$, consistently estimates the factor vector γ_t in a sense made precise in Section 4.2.

Step three treats the estimated factor vectors $\{\hat{\gamma}_t : t \in [T]\}$ as data to construct regression-based imputations of missing outcomes. We regress observed outcomes on the estimated factor vectors with unit-specific slopes $\hat{\lambda}_i$, outcome fixed effects $\hat{\gamma}_{t0}$, and covariates X_{it} .²¹

$$(\hat{\gamma}_0, \hat{\beta}, \{\hat{\lambda}_i\}_{i=1}^N) := \arg \min_{\hat{\gamma}_0 \in \mathbb{R}^T, \hat{\beta} \in \mathbb{R}^q, \hat{\lambda}_i \in \mathbb{R}^r} \left\{ \frac{1}{N} \sum_{i=1}^N \sum_{t \in \mathcal{T}_{C_i}} (Y_{it} - [\hat{\gamma}'_t \hat{\lambda}_i + \hat{\gamma}_{t0} + X'_{it} \hat{\beta}])^2 : \hat{\Gamma}' \hat{\gamma}_0 = \mathbf{0}_r \right\}. \quad (5)$$

We then estimate the mean μ_{ct} of any observed or counterfactual outcome t for cohort c by averaging asymptotically unbiased imputations of that outcome for the units in that cohort, where $N_c := \sum_{i=1}^N \mathbb{1}\{C_i = c\}$ denotes the number of units in cohort c .²²

$$\hat{\mu}_{ct} := \frac{1}{N_c} \sum_{i=1}^N \mathbb{1}\{C_i = c\} [\hat{\gamma}'_t \hat{\lambda}_i + \hat{\gamma}_{t0} + X'_{it} \hat{\beta}], \quad \hat{\mu}_c := (\hat{\mu}_{c1}, \dots, \hat{\mu}_{cT})'. \quad (6)$$

²⁰Recall that eigenvectors are unique up to magnitudes, signs, and permutations between indices corresponding to repeated eigenvalues.

²¹To implement this constrained regression using standard software, the outcome fixed effect $\hat{\gamma}_{t0}$ can be replaced by including the t -th row of any $T \times (T - r)$ matrix of eigenvectors corresponding to $\hat{A}$'s $T - r$ largest eigenvalues as additional regressors; the inner product of this vector with its coefficient vector will equal $\hat{\gamma}_{t0}$.

²²These regression-based imputations can also be used for downstream analyses that accommodate estimation error (Arkhangelsky, Yanagimoto, and Zohar, 2025; Borusyak et al., 2024; Kline, 2024; Kline, Saggio, and Sölvsten, 2020).

The constraint $\sum_{t=1}^T \hat{\gamma}_{t0} \hat{\gamma}_t = \mathbf{0}_r$ in (5) is one of many equivalent normalizations that ensure $\hat{\gamma}_0$ is uniquely defined without affecting $\hat{\mu}_{ct}$. When $\hat{\gamma}_t = 1$, the regression (5) and estimator (6) reduce to a TWFE model-based imputation estimator like those studied in Jochmans and Weidner (2019) and Borusyak et al. (2024), as discussed in Section 2.2.²³

As discussed in Section 2.1, estimands of interest are more often known functions $\theta = h(\mu, \eta)$ of cohort outcome means $\mu := (\mu'_1, \dots, \mu'_C)'$ and estimable nuisance parameters η . Examples include the dynamic treatment effects for event studies in Example D.1 of Appendix D and the match outcome attribution statistics in Example 1. Extending step three, we compute a plug-in estimate $\hat{\theta} := h(\hat{\mu}, \hat{\eta})$, where $\hat{\mu} := (\hat{\mu}'_1, \dots, \hat{\mu}'_C)'$ collects the estimates from (6) across cohorts and $\hat{\eta}$ estimates η .

In step four, given a size α , we construct simultaneous $1 - \alpha$ confidence intervals for the p coordinates of θ using a Bayesian bootstrap (Rubin, 1981).²⁴ Since our bootstrap algorithm is similar to other simultaneous inference procedures (see Chernozhukov, Fernández-Val, and Melly (2013), for example), we defer a detailed description to Appendix D.2 for brevity.

4 Theoretical Properties

4.1 Factor Identification Given Cohort-Specific Factors

To highlight our contributions, this section takes the cohort-specific factor projection matrices $\Pi(E_c \Gamma)$ as given. Section 4.3 discusses assumptions from the factor model literature on short, balanced panels under which $\Pi(E_c \Gamma)$ are identified, meaning they can be expressed as functions of each cohort c 's observed data distribution.

To characterize when factor vectors for all outcomes can be recovered with respect to a common basis given the cohort-specific factor projection matrices $\Pi(E_c \Gamma)$, we introduce the *Observed Outcome Overlap (O^3) Algorithm*, defined formally in Algorithm 2. The algorithm first constructs an O^3 Graph with one vertex per cohort and an undirected edge between cohorts sharing at least r observed outcomes. It then computes the O^3 Graph's connected components and creates a *super cohort* for each, defined as the union of cohorts in that connected component.

The algorithm then repeats this process with super cohorts playing the role of the original cohorts: it constructs a new O^3 Graph with super cohorts as vertices, where a super cohort $\mathcal{S} \subseteq [C]$'s observed outcome set $\mathcal{T}_{\mathcal{S}} := \cup_{c \in \mathcal{S}} \mathcal{T}_c$ is the union of its constituent cohorts $c \in \mathcal{S}$; then it computes connected components and merges super cohorts accordingly. The algorithm terminates when no further merging occurs, which is guaranteed within a finite number of iterations K .²⁵ Let $M^{(k)}$ denote the number

²³This normalization simplifies our theoretical results in Section 4. When $\hat{\gamma}_t = 1$, the normalization of $\hat{\gamma}_0$ in (5) is equivalent to the commonly imposed normalization that outcome fixed effect estimates sum to zero in TWFE regressions.

²⁴Simultaneous coverage $1 - \alpha$ means all coordinates of θ lie inside their respective intervals with probability at least $1 - \alpha$. Any bootstrap procedure that weights or resamples units could be used, but we prefer weighted procedures with continuously distributed weights to avoid cases where no units from some cohort are sampled.

²⁵Across iterations, the number of connected components weakly decreases and is always at least one; it cannot decrease further once super cohorts stop changing.

Algorithm 2: Observed Outcome Overlap (O^3) Algorithm

Input: observed outcomes per cohort $\{\mathcal{T}_c\}_{c=1}^C$, factor model rank r

- 1 Initialize initial super cohorts $\mathcal{S}_c^{(0)} := \{c\}$ for $c \in [C]$, set $M^{(0)} := C$
 repeat for $k = 1, 2, \dots$
 - 2 New undirected O^3 Graph $\mathcal{G}_r^{(k)}$ with $M^{(k-1)}$ vertices, one per super cohort
 - 3 Add edge in $\mathcal{G}_r^{(k)}$ between vertices m_1 and m_2 if $|\mathcal{T}_{\mathcal{S}_{m_1}^{(k-1)}} \cap \mathcal{T}_{\mathcal{S}_{m_2}^{(k-1)}}| \geq r$
 - 4 Compute $M^{(k)}$ connected components (CCs) of $\mathcal{G}_r^{(k)}$: $\{\mathcal{M}_m^{(k)}\}_{m=1}^{M^{(k)}} \subseteq [M^{(k-1)}]$
 - 5 k super cohorts union same-CC $k-1$ super cohorts: $\mathcal{S}_m^{(k)} := \bigcup_{m' \in \mathcal{M}_m^{(k)}} \mathcal{S}_{m'}^{(k-1)}$
 - until** $M^{(k)} = M^{(k-1)}$, where K denotes the final iteration count
-

of super cohorts after iteration k .

Our key requirement for recovering all factor vectors with respect to a common basis is that the O^3 Algorithm converges to a single super cohort containing all cohorts. Since the algorithm depends only on the observed outcome sets $\mathcal{T}_c$ and is computationally efficient,²⁶ researchers can verify this condition in practice. This single-super-cohort requirement implies every cohort has at least r observed outcomes: if $T_c < r$ for some cohort c , no other cohort can share r or more outcomes with c . Consequently, our identification results hold only for units with at least r observed outcomes, i.e., conditional on $T_{C_i} \geq r$.²⁷ When the algorithm converges to multiple super cohorts, our result below instead applies separately within each final super cohort and the subsets of outcomes observed for at least one cohort in those super cohorts. In Appendix F.2, we show that when $r = 1$, single-super-cohort convergence is equivalent to standard graph connectedness assumptions in the literature on fixed-effect models of bipartite network data under strict exogeneity (Abowd et al., 2002; Bonhomme, 2020; Bonhomme et al., 2019; Hull, 2018; Jochmans and Weidner, 2019; Kline, 2024).

We can now state our main identification result as follows:

Theorem 1. *Suppose the factor vectors $\{\gamma_t : t \in [T]\}$ are in general position and the O^3 Algorithm converges to a single super cohort ($\mathcal{S}_1^{(K)} = [C]$) for whom all outcomes are observed ($\mathcal{T}_{\mathcal{S}_1^{(K)}} = [T]$).²⁸ Then Γ 's column space is exactly the null space of the population APM A .*

That Γ 's column space lies in A 's null space follows directly without assumptions on factor vectors (see Footnote 17). The converse—that A 's null space lies in Γ 's column space given single-super-cohort O^3 Algorithm convergence—requires more involved arguments provided in Appendix A. The general position assumption ensures each factor affects at least one observed outcome for every cohort. While this assumption simplifies exposition and makes the O^3 Algorithm depend only on observed

²⁶A crude time complexity bound is $\mathcal{O}(C^3)$: there are $K = \mathcal{O}(C)$ iterations since at least one pair of super cohorts merges in each non-final iteration, and computing connected components is linear in vertices and edges (Cormen, Leiserson, Rivest, and Stein, 2022), with at most C vertices and $\mathcal{O}(C^2)$ edges per graph.

²⁷This is analogous to restricting samples in bipartite matching settings to row-type units matched with at least two column-type units.

²⁸A collection of vectors $\{v_j \in \mathbb{R}^d : j \in [J]\}$ is in general position if every size- d subset is linearly independent.

quantities, we can relax it by instead defining edges based directly on the factor vectors corresponding to those super cohorts' overlapping observed outcomes.²⁹

4.2 Consistency, Asymptotic Linearity Given Cohort-Specific Factors

Having identified the column space of Γ given cohort-specific factor projection matrices $\Pi(E_c\Gamma)$, we now show that the plug-in estimator from Section 3 is asymptotically linear. We first introduce some additional notation. Let $\hat{\mathbb{E}}_N$ denote expectation under the empirical measure (i.e., $\hat{\mathbb{E}}_N[V_i] := \sum_{i=1}^N V_i/N$). For square matrices M_1 , M_2 , and M_3 , define the multilinear function $g(M_1, M_2, M_3) := M_1 M_2 M_3 + M_3 M_2 M_1$. Let the T -vector $Y_i^* := (Y_{i1}^*, \dots, Y_{iT}^*)'$ collect unit i 's outcomes, let $Y_i := E_{C_i} Y_i^*$ denote observed outcomes,³⁰ and let $\varepsilon_i := (\varepsilon_{i1}, \dots, \varepsilon_{iT})'$ collect i 's outcome residuals. Finally, let $\mu_{X,c} := \mathbb{E}[X_i \mid C_i = c]$ and $\hat{\mu}_{X,c} := \hat{\mathbb{E}}_N[X_i \mid C_i = c]$ denote cohort c 's population and sample mean covariate matrices.

As in Section 4.1, we assume access to consistent estimators $\Pi(\hat{\Gamma}_c)$ of $\Pi(E_c\Gamma)$ satisfying the following asymptotic linearity condition:

Assumption 1. For each cohort $c \in [C]$, there exists a $(T \times T)$ -matrix-valued function $\phi_{\Gamma,c}$ of C_i , Y_i , and X_i such that $\mathbb{E}[\phi_{\Gamma,c}(C_i, Y_i, X_i)] = \mathbf{0}_{T \times T}$, $\mathbb{E}[\|\phi_{\Gamma,c}(C_i, Y_i, X_i)\|_{\text{op}}^2] < \infty$, and

$$\left\| \left[\Pi(\hat{\Gamma}_c) - \Pi(E_c\Gamma) \right] - \hat{\mathbb{E}}_N[\phi_{\Gamma,c}(C_i, Y_i, X_i)] \right\|_{\text{op}} = o_{\mathbb{P}}(N^{-1/2}).$$

Section 4.3 discusses low-level assumptions from the literature on short, balanced panel factor models under which estimators satisfying Assumption 1 exist. For example, with no covariates ($X_i = \mathbf{0}_{T \times q}$), the Principal Components (PC) estimator applied to cohort c 's data satisfies Assumption 1, as shown in Appendix F.3.

Under this assumption, the column space of our plug-in estimator $\hat{\Gamma}$ as represented by $\Pi(\hat{\Gamma})$ is consistent for the column space of Γ as represented by $\Pi(\Gamma)$ and asymptotically linear:

Theorem 2. Suppose that Theorem 1 and Assumption 1 hold. Then as $N \rightarrow \infty$,

$$\left\| [\Pi(\hat{\Gamma}) - \Pi(\Gamma)] - \hat{\mathbb{E}}_N[\phi_A(C_i, Y_i, X_i)] \right\|_{\text{op}} = o_{\mathbb{P}}(N^{-1/2}), \quad (7)$$

where $\Pi(\hat{\Gamma})$'s influence function

$$\phi_A(C_i, Y_i, X_i) := g\left(A^+, \frac{1}{C} \sum_{c=1}^C \phi_{\Gamma,c}(C_i, Y_i, X_i), \Pi(\Gamma)\right) \quad (8)$$

has zero mean and $\mathbb{E}[\|\phi_A(C_i, Y_i, X_i)\|_{\text{op}}^2] < \infty$.

²⁹ Specifically, we add an edge between super cohorts if the span of factor vectors for their overlapping outcomes has dimension r . Formally, replace Line 3 of Algorithm 2 with: add an edge between m_1 and m_2 if $\text{rank}(E_{\mathcal{T}_{S_{m_1}}^{(k)}} E_{\mathcal{T}_{S_{m_2}}^{(k)}} \Gamma) = r$. Single-super-cohort convergence of this ‘oracle’ algorithm implies $\text{rank}(E_c\Gamma) = r$; see Appendix F.1.

³⁰ Although Y_i has undefined entries for unobserved outcomes, all entries of $E_{C_i} Y_i$ are well-defined since those corresponding to unobserved outcomes are zero.

We prove Theorem 2 in Appendix C.1. The proof uses an exact, first-order expansion of the operator mapping a symmetric matrix to the projection onto a subset of its eigenspaces, derived via Kato's integral (Kato, 1949). Since projection matrices uniquely represent column spaces, this expansion directly bounds the second-order error of $\Pi(\hat{\Gamma})$ under minimal assumptions on A 's eigenvalues. This expansion appears to be new and may be of independent interest, so we provide a self-contained description in Appendix B.

Next, we show that $\hat{\mu}_c$ is asymptotically linear, which requires additional regularity conditions. First, we impose the following moment bounds on our model's primitives:

Assumption 2. $\mathbb{E}[\|\lambda_i\|_2^2] < \infty$, $\mathbb{E}[\|X_i\|_{\text{op}}^4] < \infty$, $\mathbb{E}[\|\varepsilon_i\|_2^2] < \infty$, and for all $c \in [C]$, $p_c > 0$.

These moment bounds in turn imply that $\mathbb{E}[\|Y_{it}^*\|_2^2] < \infty$, which is sufficient for the proof of the result below. In addition, we require an assumption to rule out collinearity between the factor vectors, outcome fixed effects, and covariates:

Assumption 3. The $(r + 1)$ -th smallest eigenvalue of the following matrix is positive.³¹

$$\mathbb{E} \left[[I - \Pi(\Gamma) \quad X_i]' \{E_{C_i} - \Pi(E_{C_i} \Gamma)\} [I - \Pi(\Gamma) \quad X_i] \right].$$

In Appendix C.2, we show that, when there are no covariates ($X_i = \mathbf{0}_{T \times q}$), Assumption 3 holds if and only if the column space of Γ equals the null space of A . As such, Assumption 3 is not necessary in this case if we assume Theorem 1 holds. We now state our main consistency and asymptotic linearity result for $\hat{\mu}_c$:

Theorem 3. Suppose that Theorem 1 and Assumptions 1, 2, and 3 hold.³² Then as $N \rightarrow \infty$,

$$\begin{aligned} \hat{\mu}_c - \mu_c &= \hat{\mathbb{E}}_N [\psi_c(C_i, Y_i, X_i)] + o_{\mathbb{P}}(N^{-1/2}), \\ \psi_c(C_i, Y_i, X_i) &:= \psi_{c, \text{reg}}(C_i, Y_i, X_i) + \psi_{c, \text{fac}}(C_i, Y_i, X_i) \end{aligned}$$

where the influence function ψ_c decomposes into two zero-mean, finite-second-moment components: $\psi_{c, \text{reg}}$ given in (C.21), the influence function if $\Pi(\Gamma)$ were known, and $\psi_{c, \text{fac}}$ given in (C.22), an additional term due to factor estimation error. Both are continuous in $\Pi(\Gamma)$, $E_c \mu_c$, $\mu_{X, c}$, and (γ_0, β) .

We defer full definitions of $\psi_{c, \text{reg}}$ and $\psi_{c, \text{fac}}$ to Theorem C.4 and prove the result in Appendix C.3.

Given $\hat{\mu}_c$'s asymptotic linearity, the asymptotic properties of the plug-in estimator $\hat{\theta}$ of a target estimand $\theta = h(\mu, \eta)$ follow easily as corollaries, where h is smooth and η are nuisance parameters that are consistently estimable at parametric rates. In particular, we prove that $\hat{\theta}$ is asymptotically normal in Appendix D.3.

When the cohort-specific factor estimator influence functions $\phi_{\Gamma, c}$ are known, ψ_c and thus the influence function for $\hat{\theta}$ from (D.6) can be consistently estimated by plugging the consistent estimators

³¹It is easy to verify this matrix is positive semi-definite.

³²In Appendix C, we prove Theorem 3 under a weaker assumption than the model (1), namely that $\mathbb{E}[\varepsilon_{it} \mid C_i, X_{it}] = 0$. However, an assumption like mean-independence of ε_{it} from C_i , λ_i , and the full collection of covariates X_i in addition is often necessary for cohort-specific factor estimators to satisfy Assumption 1; see Section 4.3.

$\Pi(\hat{\Gamma})$, $E_c \hat{\mu}_c$, $\hat{\mu}_{X,c}$, and $(\hat{\gamma}_0, \hat{\beta})$ into $\phi_{\Gamma,c}$ for their corresponding population versions, since ψ_c is smooth in these quantities. As such, one could use a plug-in procedure for valid inference in principle. However, deriving $\phi_{\Gamma,c}$ for every cohort-specific factor estimator in Section 4.3 is prohibitively laborious. To maintain flexibility in the choice of cohort-specific factor estimators, we opt for our Bayesian bootstrap approach. The resulting simultaneous confidence intervals from Section D.2 achieve valid coverage of θ under minimal regularity conditions without requiring knowledge of the cohort-specific influence functions $\phi_{\Gamma,c}$, as proved in Appendix D.3.

4.3 Identifying and Estimating Cohort-Specific Factors

We now turn to discussing how to identify and estimate the cohort-specific factor projection matrices $\Pi(E_c \Gamma)$ since, without additional assumptions, $\Pi(E_c \Gamma)$ cannot be identified (Anderson and Rubin, 1956). Below, we discuss several common assumptions from the literature on estimating factor models with no missing data that enable identification and estimation of $\Pi(E_c \Gamma)$ when T_c remains fixed as N_c grows. Our APM approach can wrap around any of these cohort-specific factor vector estimators.

Models with uncorrelated errors across outcomes. When errors ε_{it} are uncorrelated across outcomes t —a common assumption in the bipartite match outcome literature (Kline et al., 2020)—the cohort-specific factor matrix can be identified without auxiliary data. In the simplest case with no covariates and homoskedastic errors across outcomes (but not necessarily across units), the canonical Principal Components (PC) estimator $\Pi(\hat{\Gamma}_{c,PC})$ yields $N_c^{1/2}$ -consistent estimates of $\Pi(E_c \Gamma)$ provided T_c remains finite as N_c grows and the loadings λ_i for units in cohort c have a full-rank covariance matrix (J. Bai, 2003; Connor and Korajczyk, 1986; Westerlund, 2020). In Appendix F.3, we prove that $\Pi(\hat{\Gamma}_{c,PC})$ is asymptotically linear, as required by Assumption 1, under slightly weaker conditions than typically imposed, using our first-order expansion of the eigenspace projection operator from Appendix B.

When errors are heteroskedastic, the PC estimator is generally inconsistent in short panels due to the incidental parameter problem (Ahn, Lee, and Schmidt, 2001; J. Bai, 2003; Chamberlain and Rothschild, 1983; Connor and Korajczyk, 1986; Lancaster, 2000; Neyman and Scott, 1948). Unlike the homoskedastic case where $T \geq r + 1$ suffices for identification, Anderson and Rubin (1956) show that $(T - r)^2 \geq T + r$ is necessary. They propose estimators requiring maximization of highly nonconcave objectives (see Anderson (1963) and Chapter 14 of Anderson (2009)). While global maximizers yield asymptotically linear estimators of $\Pi(E_c \Gamma)$, off-the-shelf algorithms are not guaranteed to converge to them, and algorithms with global optimality guarantees are difficult to scale even for moderate r and T_c (Bertsimas, Copenhaver, and Mazumder, 2017; Khamaru and Mazumder, 2019).

Another line of work develops computationally efficient estimators based on internal instruments, proposed by Madansky (1964) and generalized by Freyberger (2018); Harding, Lamarche, and Muris (2025); Heckman and Scheinkman (1987). These methods assume $T \geq 2r + 1$ and accommodate covariates, but typically only produce aligned factor estimates for subsets of factors. An exception

is Hägglund (1982), which yields an $N_c^{1/2}$ -consistent and asymptotically linear estimate of the entire factor matrix column space under the stronger assumption that $T \geq 3r$.

Models with correlated errors and auxiliary data. With auxiliary data, we can relax the assumption of uncorrelated errors across outcomes. One approach uses quasi-differencing to project out the factor component (Chamberlain, 1984). With sufficiently many time-varying covariates, the GMM estimators of Ahn et al. (2001); Ahn, Lee, and Schmidt (2013); Chamberlain (1984, 1992); Phillips and Williams (2024); Robertson and Sarafidis (2015) are $N_c^{1/2}$ -consistent and asymptotically linear in short panels with heteroskedastic and correlated errors. Other $N_c^{1/2}$ -consistent methods using covariates include Projected Principal Component Analysis (Fan, Liao, and Wang, 2016) and Diversified Projections (Duan, Pelger, and Xiong, 2023). Alternatively, if some covariates share a low-rank structure with the outcomes,³³ one can apply Common Correlated Effects (Pesaran, 2006; Westerlund, Petrova, and Norkute, 2019), Essential Regression (Bing, Bunea, and Wegkamp, 2022), or transfer learning (Duan et al., 2023) to estimate $\Pi(E_c\Gamma)$.

5 Empirical Illustration

We evaluate our procedure as an alternative to TWFE model-based methods for decomposing observed wage differences across locations into causal location effects versus worker sorting, in the spirit of Card et al. (2025). As discussed in Example 1, such analyses require predicting counterfactual wages for workers at locations where they are not observed. Unlike TWFE-based imputations, which rule out complementarities and multidimensional match effects (Section 2.2), our factor model-based method accommodates these features.³⁴

5.1 Setting and Sample Construction

Our empirical illustration uses the Veneto Worker Histories (VWH) dataset derived from Italian social security administration data on the weekly wages earned across different firms of every person who ever lived or worked in any of the seven Veneto provinces from 1975 to 2001.³⁵ Each observation records a worker-firm-year triple, including weeks worked in the year and total pay.

To construct our panel, we restrict to firms existing in Veneto from 1998 through 2001 and workers employed by those firms during that period. We then cluster firms within each province into three

³³In this case, one could alternatively treat the covariates as additional, always-observed outcomes, creating overlapping observed “outcomes” between every pair of cohorts, strengthening identification of $\Pi(\Gamma)$.

³⁴Bonhomme et al. (2019) note that typical empirical tests claiming no match effects have no power under models with worker-firm complementarities (see Card et al. (2013) for an example). Card et al. (2013) and Woodcock (2015) find evidence for match effects using repeated measurements of the same matches over time, though such estimates are noisy and/or require random-effects-like assumptions. Bonhomme et al. (2019) estimate a model with discrete worker and firm types and also find evidence of complementarities.

³⁵The VWH dataset was developed by the Economics Department in Università Ca’ Foscari Venezia under the supervision of Giuseppe Tattara, and can be accessed at <https://www.frdp.org/en/dati/dati-inps-carriere-lavorative-in-veneto/>. The code for our empirical illustration is available at <https://github.com/brad-ross/apm-paper>.

with improved statistical efficiency, though we leave fully developing such refinements to future work.

One straightforward extension assigns more general positive weights w_c to cohorts and constructs a weighted APM:

$$A(w) := \sum_{c=1}^C w_c (E_c - \Pi(E_c \Gamma)),$$

where the original APM is recovered by setting $w_c = 1/C$ for all c . By an argument similar to the proof of Theorem 1, the null space of any weighted APM equals the column space of Γ provided the O^3 Algorithm converges to a single super cohort. If empirical weight analogs $\hat{w}_c$ are $N^{1/2}$ -consistent and asymptotically linear, we can modify the proof of Theorem 3 to derive an influence function for the weighted analog of $\hat{\mu}_c$ that additionally depends on the influence functions of the weights.

We can also expand the class of APMs substantially by considering polynomials of $E_c - \Pi(\hat{\Gamma}_c)$:

$$A_{\text{poly}}(w) = \sum_{k \geq 1} \sum_{(c_1, \dots, c_k) \in [C]^k} w_{c_1, \dots, c_k} \prod_{\ell=1}^k (E_{c_\ell} - \Pi(E_{c_\ell} \Gamma)), \quad w_c > 0, \quad c \in [C].$$

for non-negative weights $w_{c_1, \dots, c_k}$ that can be estimated at parametric rates. Since $(E_c - \Pi(E_c \Gamma))\Gamma = 0$ for all c , Γ 's column space lies in $A_{\text{poly}}(w)$'s null space. Moreover, $A_{\text{poly}}(w)$ is lower-bounded by $A(w_{k=1})$ in the positive semi-definite order (where $w_{k=1}$ is the subset of w for $k = 1$), so its $(r + 1)$ -th eigenvalue is bounded away from zero so long as $A(w_{k=1})$'s is. As such, its null space must be exactly Γ 's column space, as implied by Theorem 1. Asymptotic linearity of polynomial APM-based estimators should follow via the Delta method, though the resultant estimators will have more complex influence functions.

Finally, we can add additional cohorts to the APM formed by merging existing cohorts and retaining their overlapping outcomes. For example, if cohort 1 observes outcomes $\{1, 2, 3\}$ and cohort 2 observes $\{2, 3, 4\}$, we can define a combined cohort $(1, 2)$ containing all units from both cohorts with observed outcomes $\{2, 3\}$. Adding these cohorts preserves identification since it can only strengthen cohort connectedness. The resulting estimators should be consistent and asymptotically linear, potentially with improved efficiency from larger combined cohorts enabling more precise cohort-specific factor estimates. However, one would have to carefully account for the statistical dependence among cohort-specific factor estimators these additional cohorts introduce.

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A Proof of Theorem 1

First, as in Footnote 17, we show that for any cohort c , the columns of Γ lie in the null space of $E_c - \Pi(E_c\Gamma)$, in which case $\text{col}(\Gamma) \subseteq \text{null}(A)$, where $\text{col}(\Gamma)$ denotes the column space of Γ . Since $E_c^2 = E_c$ and $E_c\Pi(E_c\Gamma) = \Pi(E_c\Gamma)E_c = \Pi(E_c\Gamma)$, we have that

$$(E_c - \Pi(E_c\Gamma))\Gamma = (E_c^2 - E_c\Pi(E_c\Gamma)E_c)\Gamma = E_c \cdot (I - \Pi(E_c\Gamma))E_c\Gamma = \mathbf{0}_{T \times r},$$

as required.

Next, we show that $\text{null}(A) \subseteq \text{col}(\Gamma)$ also holds. Define the matrix

$$\tilde{A} := \frac{1}{\sqrt{C}} \begin{bmatrix} E_1 - \Pi(E_1\Gamma) \\ \vdots \\ E_C - \Pi(E_C\Gamma) \end{bmatrix},$$

and note that since

$$(E_c - \Pi(E_c\Gamma))^2 = E_c^2 - E_c\Pi(E_c\Gamma) - \Pi(E_c\Gamma)E_c - \Pi(E_c\Gamma)^2 = E_c - \Pi(E_c\Gamma)$$

and $E_c - \Pi(E_c\Gamma)$ is symmetric, $A = \tilde{A}'\tilde{A}$, implying that $\text{null}(A) = \text{null}(\tilde{A})$.⁴³

Consider any $v \in \text{null}(\tilde{A})$. We will show by induction on iterations $k \geq 0$ of the O^3 Algorithm (Algorithm 2) that for any super cohort $m \in \{1, \dots, M^{(k)}\}$, $E_{\mathcal{T}_{\mathcal{S}_m^{(k)}}} v \in \text{col}(E_{\mathcal{T}_{\mathcal{S}_m^{(k)}}}\Gamma)$ (recall from Section 2.3 that, for any subset of outcomes $\mathcal{T} \subseteq [T]$, $E_{\mathcal{T}}$ is defined as the diagonal matrix whose t -th diagonal entry is one if $t \in \mathcal{T}$ and zero otherwise). As the base case, take $k = 0$. By the definitions of v and $\tilde{A}$, for any cohort c , $(E_c - \Pi(E_c\Gamma))v = \mathbf{0}_T$, which in turn implies that, since the initial super cohorts $\mathcal{S}_c^{(0)}$ are singletons corresponding to cohorts c :

$$E_{\mathcal{T}_{\mathcal{S}_c^{(0)}}} v = E_c v = \Pi(E_c\Gamma)v = \Pi(E_{\mathcal{T}_{\mathcal{S}_c^{(0)}}}\Gamma)E_{\mathcal{T}_{\mathcal{S}_c^{(0)}}} v.$$

Therefore, $E_{\mathcal{T}_{\mathcal{S}_c^{(0)}}} v \in \text{col}(E_{\mathcal{T}_{\mathcal{S}_c^{(0)}}}\Gamma)$, as required.

For the inductive step, consider $k \geq 1$ and suppose that for all super cohorts $m \in \{1, \dots, M^{(k-1)}\}$, we have that $E_{\mathcal{T}_{\mathcal{S}_m^{(k-1)}}} v \in \text{col}(E_{\mathcal{T}_{\mathcal{S}_m^{(k-1)}}}\Gamma)$. Then for every iteration- $(k-1)$ super cohort m , there exists a vector $\omega_m \in \mathbb{R}^r$ such that $E_{\mathcal{T}_{\mathcal{S}_m^{(k-1)}}} v = E_{\mathcal{T}_{\mathcal{S}_m^{(k-1)}}}\Gamma\omega_m$. Consider any edge (m_1, m_2) in the iteration- k O^3 Graph $\mathcal{G}_r^{(k)}$. Since diagonal matrices commute, we have that

$$\begin{aligned} E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(k-1)}}} E_{\mathcal{T}_{\mathcal{S}_{m_2}^{(k-1)}}} v &= E_{\mathcal{T}_{\mathcal{S}_{m_2}^{(k-1)}}} E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(k-1)}}} v = E_{\mathcal{T}_{\mathcal{S}_{m_2}^{(k-1)}}} E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(k-1)}}}\Gamma\omega_{m_1} = E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(k-1)}}} E_{\mathcal{T}_{\mathcal{S}_{m_2}^{(k-1)}}}\Gamma\omega_{m_1}, \\ E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(k-1)}}} E_{\mathcal{T}_{\mathcal{S}_{m_2}^{(k-1)}}} v &= E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(k-1)}}} E_{\mathcal{T}_{\mathcal{S}_{m_2}^{(k-1)}}}\Gamma\omega_{m_2}, \end{aligned}$$

implying that $E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(k-1)}}} E_{\mathcal{T}_{\mathcal{S}_{m_2}^{(k-1)}}}\Gamma(\omega_{m_1} - \omega_{m_2}) = \mathbf{0}_T$. By Line 3 in Algorithm 2, there must be at least r overlapping observed outcomes between super cohorts m_1 and m_2 , so since the factor vectors γ_t are in general position, $E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(k-1)}}} E_{\mathcal{T}_{\mathcal{S}_{m_2}^{(k-1)}}}\Gamma$ must have full rank. This observation in turn implies that the null space of $E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(k-1)}}} E_{\mathcal{T}_{\mathcal{S}_{m_2}^{(k-1)}}}\Gamma$ is trivial and $\omega_{m_1} - \omega_{m_2}$ lies in it, meaning $\omega_{m_1} = \omega_{m_2}$. Now consider any connected component $\mathcal{M}_{m'}^{(k)}$ of $\mathcal{G}_r^{(k)}$, which corresponds to super cohort $\mathcal{S}_{m'}^{(k)}$. Since all iteration- $(k-1)$ super cohorts in the connected component $\mathcal{M}_{m'}^{(k)}$ of $\mathcal{G}_r^{(k)}$ must be connected by some sequence of edges in $\mathcal{G}_r^{(k)}$, chaining the edge-wise ω_m -equality logic above along that sequence implies

⁴³ It is straightforward to check that if $v \in \text{null}(\tilde{A})$, then $v \in \text{null}(A) = \text{null}(\tilde{A}'\tilde{A})$. To check the other direction of inclusion, for any $v \in \text{null}(A)$, we have that $\mathbf{0} = Av = \tilde{A}'\tilde{A}v$. Further, we have that $\mathbf{0} = v'\mathbf{0} = v'\tilde{A}'\tilde{A}v = (\tilde{A}v)'(\tilde{A}v)$, so it must be that $\tilde{A}v = \mathbf{0}$, as required.

that there must exist some common vector $\omega \in \mathbb{R}^r$ such that $\omega_m = \omega$ for all $m \in \mathcal{M}_{m'}^{(k)}$, meaning that $E_{\mathcal{T}_{\mathcal{S}_m^{(k-1)}}} v = E_{\mathcal{T}_{\mathcal{S}_m^{(k-1)}}} \Gamma \omega$ for all prior-iteration super cohorts $q \in \mathcal{Q}_{q'}^{(k)}$.

Now consider any observed outcome t for the iteration- k super cohort $\mathcal{S}_{m'}^{(k)}$, i.e. $t \in \mathcal{T}_{\mathcal{S}_{m'}^{(k)}}$. Note that this outcome must also be observed for at least one iteration- $(k-1)$ super cohort m_t belonging to the connected component $\mathcal{M}_{m'}^{(k)}$, i.e. $t \in \mathcal{T}_{\mathcal{S}_{m_t}^{(k-1)}}$. Letting v_t denote the t -th entry of v and e_t denote the t -th standard basis vector, we have that

$$v_t = e_t' E_{\mathcal{T}_{\mathcal{S}_{m_t}^{(k-1)}}} \Gamma \omega = \gamma_t' \omega. \quad (\text{A.1})$$

Stacking (A.1) across $t \in \tilde{\mathcal{T}}_{q'}^{(k)}$, we have that $E_{\mathcal{T}_{\mathcal{S}_{m'}^{(k)}}} v = E_{\mathcal{T}_{\mathcal{S}_{m'}^{(k)}}} \Gamma \omega$, meaning that $E_{\mathcal{T}_{\mathcal{S}_{m'}^{(k)}}} v \in \text{col}(E_{\mathcal{T}_{\mathcal{S}_{m'}^{(k)}}} \Gamma)$, as required.

Finally, recall that K denotes the number of iterations required for the O^3 Algorithm to converge and that we assumed the final number of super cohorts was $M^{(K)} = 1$. Since all super cohorts must be unions of prior-iteration super cohorts and every outcome is observed for some initial super cohort (since every outcome is observed for at least one cohort), it must be that this single super cohort is the set of all cohorts, i.e. $\mathcal{S}_1^{(K)} = [C]$, and therefore that all outcomes are observed for this final super cohort, i.e. $\mathcal{T}_{\mathcal{S}_1^{(K)}} = [T]$. As such, the inductive logic above implies that

$$v = E_{\mathcal{T}_{\mathcal{S}_1^{(K)}}} v \in \text{col}(E_{\mathcal{T}_{\mathcal{S}_1^{(K)}}} \Gamma) = \text{col}(\Gamma).$$

Therefore, $\text{null}(\tilde{A}) = \text{null}(A) \subseteq \text{col}(\Gamma)$, as required.

B An Eigenspace Operator Exact First-Order Expansion

In this section, we derive an exact, first-order expansion of the operator mapping a symmetric matrix into the projection matrix onto the space spanned by some subset of its eigenvectors. Our expansion is based on Kato's integral, which characterizes the projection matrix onto the space spanned by some subset of a matrix's eigenvectors as a contour integral of that matrix's resolvent [Kato \(1949\)](#). We explicitly characterize the approximation error in our expansion including exact constants, unlike the asymptotic expansions given in e.g. [Kato \(1980\)](#) and [J.-g. Sun \(1991\)](#) (which is applied in [Simons \(2023\)](#) to construct hypothesis tests concerning eigenspaces). As such, our result below may be of independent interest.

To describe our expansion, we first introduce some notation. Let $\mathbb{S}^d$ denote the set of real-valued, d -dimensional symmetric matrices, and for any $M \in \mathbb{S}^d$, let $\lambda_j(M)$ be the j -th smallest eigenvalue of A , where

$$\lambda_0(M) := -\infty < \lambda_1(M) \leq \dots \leq \lambda_d(M) < \lambda_{d+1}(M) := \infty,^{44}$$

⁴⁴The eigenvalues of any symmetric matrix are all real-valued.

$$\begin{aligned}
& + o_{\mathbb{P}}(N^{-1/2}) \\
& = \hat{\mathbb{E}}_N[\psi_{c,i,\text{reg}}] + \hat{\mathbb{E}}_N[\psi_{c,i,\text{fac}}] + o_{\mathbb{P}}(N^{-1/2})
\end{aligned}$$

The moment conditions on ψ_i are directly consequences of Lemma C.13 and Assumption 2.

By (C.13) in Lemma C.2,

$$H_0 A_{0,\omega}^{-1} = H_0 A_0^+.$$

Thus, we can also replace the term $H_0 A_{0,\omega}^{-1}$ by $H_0 A_0^+$ in the above expression. For any fixed $\omega > 0$, Lemma C.2 implies $A_{0,\omega}$ is invertible. Since $A_{0,\omega}$ is continuous in $\Pi(\Gamma)$, $A_{0,\omega}^{-1}$ is also continuous in $\Pi(\Gamma)$.

D Target Parameters: Examples, Estimation, and Inference

In this appendix, we provide several examples of empirically relevant target parameters θ that researchers can estimate and conduct inference on using our estimator. Then, we describe our Bayesian-bootstrap-based simultaneous inference procedure for θ in detail. Finally, we establish that our plug-in estimator $\hat{\theta}$ described in Section 3 is a consistent and asymptotically normal estimator of θ , and that the Bayesian-bootstrap-based simultaneous confidence intervals described in Section D.2 have valid simultaneous coverage.

D.1 Examples of Target Parameters

Here, we introduce two examples of target parameters $\theta = h(\mu, \eta)$ that aggregate cohort outcome means μ and nuisance parameters η nonlinearly through h as introduced in Section 2.1:

Example D.1 (Dynamic Treatment Effects). In event study settings, researchers are often interested in reporting dynamic treatment effect paths, i.e. average effects of a treatment across different numbers of time periods relative to units' treatment times. Recall that, in our notation, outcome Y_{it}^* refers to unit i 's potential outcome in period t had they not yet been treated by period t (often denoted by $Y_{it}(\infty)$ in this literature), C_i refers to unit i 's first treatment period, and the set of observed outcomes for the units treated in period c is $\mathcal{T}_c \subset \{1, \dots, c-1\}$. To express dynamic treatment effects in the form $h(\mu, \eta)$, we first define Z_{it}^* as unit i 's potential outcome in period t had they been treated in period C_i , and, to distinguish between observed and missing outcomes, we let $Z_{it} = Z_{it}^*$ if unit i was treated by period t (i.e. $C_i \leq t$) and $Z_{it} = \emptyset$ otherwise.

Assuming we observe either Y_{it}^* or Z_{it}^* in every period for simplicity,⁵⁰ we can write a coordinates of the p -dimensional parameter vector whose entries correspond to the dynamic treatment effects from

⁵⁰In many settings like Figure 1a's, we do not observe control or treated potential outcome in every period. Callaway and Sant'Anna (2021) and L. Sun and Abraham (2021) discuss several ways of defining estimands that take this additional unbalancedness into account; for brevity, we simply note that these estimands can also be written in the form $h(\mu, \eta)$ and refer the interested reader to Section 3.1.1 of Callaway and Sant'Anna (2021) for details.

b periods before treatment through $p - b$ treated periods in the form of Equation (3.4) in [Callaway and Sant'Anna \(2021\)](#) or Equation (26) in [L. Sun and Abraham \(2021\)](#):

$$\theta_{\text{dyn},j} := \sum_{t=1}^T \sum_{c=1}^C \mathbb{1}\{t - c = j - b - 1\} p_c \cdot \left(\underbrace{\mathbb{E}[\mathbb{1}\{t \geq c\} Z_{it} + \mathbb{1}\{t < c\} Y_{it} \mid C_i = c]}_{m_{ct}} - \mu_{ct} \right), \quad (\text{D.1})$$

where $j \in \{1, \dots, p\}$ indexes the coordinates of θ_{dyn} , $t - c$ reflects the index of the current period relative to cohort c 's treatment time (with zero corresponding to a cohort's first treated period in relative time), and m_{ct} denotes the average observed outcome for units in cohort c in period t ; when period t is before cohort c 's treatment time c , the observed outcome is Y_{it} which equals the control potential outcome Y_{it}^* , and when period t is after cohort c 's treatment time c , the observed outcome is Z_{it} which equals the treated potential outcome Z_{it}^* . We note that $\theta_{\text{dyn},j}$ can be defined without any restrictions on heterogeneity in treatment effects $Z_{it}^* - Y_{it}^*$ across units or time periods ([Callaway and Sant'Anna, 2021](#); [L. Sun and Abraham, 2021](#)).

Example 1 (continued). We continue our discussion of the nonparametric decomposition of differences in observed outcome means introduced in Example 1. We note that the definitions of $\theta_{\text{col},t_1,t_2}$ and $\theta_{\text{row},t_1,t_2}$ from Example 1 are only defined in terms of outcomes, which is beneficial for interpretation ([Hull, 2018](#); [Kline, 2024](#)), but they coincide exactly with S_{place} and S_{pat} in [Finkelstein et al. \(2016\)](#) when Y_{it}^* is determined by the TWFE model (2):

$$\begin{aligned} \theta_{\text{col},t_1,t_2} &\propto \gamma_{t_1} - \gamma_{t_2} \\ \theta_{\text{row},t_1,t_2} &\propto \mathbb{E}[\lambda_i \mid t_1 \in \mathcal{T}_{C_i}] - \mathbb{E}[\lambda_i \mid t_2 \in \mathcal{T}_{C_i}]. \end{aligned}$$

As discussed in [Finkelstein et al. \(2016\)](#), one can also generalize this decomposition by studying differences in average observed match outcomes $\bar{\mu}_{\tilde{\mathcal{T}}_1} - \bar{\mu}_{\tilde{\mathcal{T}}_2}$ between two *groups* of outcomes $\tilde{\mathcal{T}}_1, \tilde{\mathcal{T}}_2 \subset [T]$ with outcomes assigned potentially unequal weights $w_t > 0$, i.e.

$$\bar{\mu}_{\tilde{\mathcal{T}}_1} - \bar{\mu}_{\tilde{\mathcal{T}}_2}, \quad \bar{\mu}_{\tilde{\mathcal{T}}} := \left(\sum_{c=1}^C \mathbb{P}(C_i = c) \sum_{t \in \tilde{\mathcal{T}} \cap \mathcal{T}_c} w_t \right)^{-1} \sum_{c=1}^C \mathbb{P}(C_i = c) \sum_{t \in \tilde{\mathcal{T}} \cap \mathcal{T}_c} w_t \mathbb{E}[Y_{it} \mid t \in \mathcal{T}_c].$$

In particular, one can define $\theta_{\text{col},\tilde{\mathcal{T}}_1,\tilde{\mathcal{T}}_2}$ and $\theta_{\text{row},\tilde{\mathcal{T}}_1,\tilde{\mathcal{T}}_2}$ analogously to $\theta_{\text{col},t_1,t_2}$ and $\theta_{\text{row},t_1,t_2}$:

$$\theta_{\text{col},\tilde{\mathcal{T}}_1,\tilde{\mathcal{T}}_2} := [\bar{\mu}_{\tilde{\mathcal{T}}_1} - \bar{\mu}_{\tilde{\mathcal{T}}_2}]^{-1} [\bar{\mu}_{\tilde{\mathcal{T}}_1}^* - \bar{\mu}_{\tilde{\mathcal{T}}_2}^*] =: 1 - \theta_{\text{row},\tilde{\mathcal{T}}_1,\tilde{\mathcal{T}}_2}, \quad \bar{\mu}_{\tilde{\mathcal{T}}}^* := \left(\sum_{t \in \tilde{\mathcal{T}}} w_t \right)^{-1} \sum_{t \in \tilde{\mathcal{T}}} w_t \mathbb{E}[Y_{it}^*].$$

Given a collection of disjoint subsets of outcomes $\tilde{\mathcal{T}}_1, \dots, \tilde{\mathcal{T}}_K$, we can summarize the pairwise differences in average observed match outcomes across all pairs of these outcome groups by averaging

$\theta_{\text{col}, \tilde{\mathcal{T}}_i, \tilde{\mathcal{T}}_j}$ over all pairs $(i, j) \in [K] \times [K]$, weighted by the total outcome weight corresponding to the union of each pair of outcome groups:

$$\bar{\theta}_{\text{col}} := \left(\sum_{i=1}^K \sum_{j=i+1}^K \sum_{t \in \tilde{\mathcal{T}}_i \cup \tilde{\mathcal{T}}_j} w_t \right)^{-1} \sum_{i=1}^K \sum_{j=i+1}^K \left(\sum_{t \in \tilde{\mathcal{T}}_i \cap \tilde{\mathcal{T}}_j} w_t \right) \theta_{\text{col}, \tilde{\mathcal{T}}_i, \tilde{\mathcal{T}}_j} =: 1 - \bar{\theta}_{\text{row}}.$$

In our empirical illustration in Section 5.3, we estimate $\bar{\theta}_{\text{col}}$ for subsets of outcomes corresponding to clusters of firms within the same provinces in the two-year window 2000 to 2001. For simplicity, we restrict our attention to $\theta_{\text{col}, t_1, t_2}$ and $\theta_{\text{row}, t_1, t_2}$ in our remaining discussion of the theory of conducting valid inference on target parameters.

D.2 A Bootstrap-Based Simultaneous Inference Procedure

To describe how we construct our Bayesian-bootstrap-based simultaneous confidence intervals, we introduce more notation. Given a vector of N non-negative weights $W := (W_1, \dots, W_N)'$ that sum to one, we will assume that our cohort-specific factor matrix estimators $\hat{\Gamma}_c$ as well as our nuisance parameter estimator $\hat{\eta}$ can be adapted to accommodate non-uniform sampling weights W , which we denote $\hat{\Gamma}_c(W)$ and $\hat{\eta}(W)$, respectively. Typically, when $W_1 = \dots = W_N = \frac{1}{N}$, we have that $\hat{\Gamma}_c(W) = \hat{\Gamma}_c$ and $\hat{\eta}(W) = \hat{\eta}$.

Next, we construct an APM using these weighted, estimated cohort-specific factor matrices $\hat{\Gamma}_1(W), \dots, \hat{\Gamma}_C(W)$:

$$\hat{A}(W) := \frac{1}{C} \sum_{c=1}^C \left[E_c - \Pi(E_c \hat{\Gamma}_c(W)) \right],$$

and we let $\hat{\Gamma}(W)$ denote the eigenvectors of $\hat{A}(W)$ corresponding to its r smallest eigenvalues. We can then define a weighted analog of the regression-based imputation estimator (5):

$$\begin{aligned} & \left(\hat{\gamma}_0(W), \hat{\beta}(W), \{\hat{\lambda}_i(W)\}_{i=1}^N \right) \\ &:= \arg \min_{\hat{\gamma}_0 \in \mathbb{R}^T, \hat{\beta} \in \mathbb{R}^q, \hat{\lambda}_i \in \mathbb{R}^r} \left\{ \sum_{i=1}^N W_i \sum_{t \in \mathcal{T}_{C_i}} (Y_{it} - [\hat{\gamma}_t(W)' \hat{\lambda}_i + \hat{\gamma}_{t0} + X'_{it} \hat{\beta}])^2 : \hat{\Gamma}'(W) \hat{\gamma}_0 = \mathbf{0}_r \right\}, \end{aligned}$$

so we can construct a weighted analog of the regression-based imputation estimator (5) as follows:

$$\begin{aligned} \hat{\mu}_{ct}(W) &:= \sum_{i=1}^N \frac{\mathbb{1}\{C_i = c\} W_i}{\sum_{j=1}^N \mathbb{1}\{C_j = c\} W_j} \left[\hat{\gamma}_t(W)' \hat{\lambda}_i(W) + \hat{\gamma}_{t0}(W) + X'_{it} \hat{\beta}(W) \right], \\ \hat{\mu}_c(W) &:= (\hat{\mu}_{c1}(W), \dots, \hat{\mu}_{cT}(W))'. \end{aligned} \tag{D.2}$$

A weighted version of our plug-in estimator for $\hat{\theta}$ is $\hat{\theta}(W) := h(\hat{\mu}(W), \hat{\eta}(W))$, where $\hat{\mu}(W)$ is the weighted analog of $\hat{\mu}$. Before continuing, we note that, again, under uniform weights $W_1 = \dots = W_N = w > 0$, all of the quantities defined previously in this paragraph equal their unweighted

counterparts.

Given this notation, our inference procedure proceeds as follows. First, for each iteration m of a large number M of repetitions,⁵¹ we take N i.i.d. draws $\xi_{m1}, \dots, \xi_{mN}$ from the Exponential(1) distribution,⁵² construct a vector $W_m := (W_{m1}, \dots, W_{mN})'$ of N normalized weights $W_{mi} := \xi_{mi} / \sum_{j=1}^N \xi_{mj}$, and compute a weighted target parameter estimate $\hat{\theta}_m^* := \hat{\theta}(W_m)$.⁵³ Next, for each coordinate $j \in \{1, \dots, p\}$ of θ , we compute an estimate $\hat{\sigma}_j$ of the standard error of $\hat{\theta}_j$, e.g.

$$\hat{\sigma}_j := \frac{q_{0.75}(\hat{\theta}_{1j}^*, \dots, \hat{\theta}_{Mj}^*) - q_{0.25}(\hat{\theta}_{1j}^*, \dots, \hat{\theta}_{Mj}^*)}{q_{0.75}(Z) - q_{0.25}(Z)}, \quad (\text{D.3})$$

where we let $q_\zeta(x_1, \dots, x_M)$ denote the ζ th quantile across scalars $x_1, \dots, x_M$, and we let $q_\zeta(Z)$ denote the ζ th quantile of the standard Gaussian distribution.⁵⁴ Finally, we let $\hat{q}_{1-\alpha}$ denote the following estimated critical value:

$$\hat{q}_{1-\alpha} := q_{1-\alpha}(z_1^*, \dots, z_M^*), \quad z_m^* := \max_{j \in \{1, \dots, p\}} \left| \hat{\theta}_{mj}^* / \hat{\sigma}_j \right|, \quad (\text{D.4})$$

and we define our simultaneous $1 - \alpha$ confidence intervals $\hat{\mathcal{C}}_j$ for θ_j across $j = 1, \dots, p$ as follows:

$$\hat{\mathcal{C}}_j := \left[\hat{\theta}_j - \hat{q}_{1-\alpha} \hat{\sigma}_j, \hat{\theta}_j + \hat{q}_{1-\alpha} \hat{\sigma}_j \right]. \quad (\text{D.5})$$

For convenience, we summarize the steps of our inference procedure in Algorithm 3.

D.3 Asymptotic Linearity, Normality, Confidence Interval Validity

Having established in Section 4.2 that $\hat{\mu}_c$ is an asymptotically linear estimator of the vector of cohort outcome means μ_c , we now establish asymptotic linearity of our plug-in estimator $\hat{\theta}$ of our target estimand $\theta = h(\mu, \eta)$ and validity of our simultaneous confidence intervals $\hat{\mathcal{C}}_j$. To do so, we assume that nuisance parameter estimator $\hat{\eta}$ is also asymptotically linear:

Assumption D.1. There exists an additional observed random vector $\varrho_i \in \mathbb{R}^m$ measurable with respect to the same probability space as (C_i, Y_i^*, X_i) that satisfies $\mathbb{E}[\varrho_i] = \mathbf{0}_m$ and $\mathbb{E}[\|\varrho_i\|_2^2] < \infty$ such

⁵¹We recommend $M \geq 500$.

⁵²Other non-negative distributions are also possible as long as they satisfy regularity conditions; see Section 3.6.2 in [van der Vaart and Wellner \(1996\)](#) for more examples.

⁵³Since T is small, each computation of $\hat{\theta}(W_m)$ should be quite fast, as discussed when describing Algorithm 1 above. Further, $\hat{\theta}(W_m)$ can be computed in parallel across iterations m , boosting computational efficiency further. However, in settings where $T \ll N$ but N and T are both very large, the simplicity of this weighted bootstrap procedure may be outweighed its computational burden. In such cases, one could instead construct a multiplier bootstrap inference procedure like the one proposed in [Belloni, Chernozhukov, Fernandez-Val, and Hansen \(2017\)](#) using the influence function expressions given in Theorem 2, Theorem 3, and Corollary D.1.

⁵⁴We suggest this interquartile-range-based estimate of estimator standard errors because it is more robust to outliers than other standard error estimators like the standard deviation over weighted bootstrap draws; see the discussion in Remark 3.2 in [Chernozhukov et al. \(2013\)](#) for details.

Algorithm 3: Bayesian Bootstrap Inference Procedure

Data: $\{(C_i, Y_i)\}_{i=1}^N$, number of bootstrap samples M .

```

1 for  $m \in \{1, \dots, M\}$  do
    Sample  $\xi_{m1}, \dots, \xi_{mN} \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(1)$ 
    Construct weight vector  $W_m := (W_{m1}, \dots, W_{mN})'$  with  $W_{mi} := \xi_{mi} / \sum_{j=1}^N \xi_{mj}$ 
    for  $c \in \{1, \dots, C\}$  do
        Compute weighted cohort outcome mean estimate vector  $\hat{\mu}_c(W_m)$  as in (D.2) (using
        Algorithm 1 with weighted analogs)
    end
    Compute weighted nuisance parameter estimates  $\hat{\eta}(W_m)$ 
    Compute weighted target parameter estimate  $\hat{\theta}_m^* = h(\hat{\mu}(W_m), \hat{\eta}(W_m))$ 
end
2 for  $j \in \{1, \dots, p\}$  do
    Compute estimate  $\hat{\sigma}_j$  of  $\hat{\theta}$ 's standard error, e.g. as in (D.3)
end
3 Compute estimated critical value  $\hat{q}_{1-\alpha}$  as in (D.4)
4 Compute simultaneous  $1 - \alpha$  confidence intervals  $\hat{\mathcal{C}}_j$  as in (D.5)
  
```

that the following expansion holds as $N \rightarrow \infty$:

$$\hat{\eta} - \eta = \hat{\mathbb{E}}_N [\varrho_i] + o_{\mathbb{P}}(N^{-1/2}).$$

In addition, we assume that the function h that defines $\theta = h(\mu, \eta)$ is sufficiently smooth:

Assumption D.2. $h: \mathbb{R}^{CT} \times \mathbb{R}^m \rightarrow \mathbb{R}^p$ is differentiable at (μ, η) .

Under the additional Assumptions D.1 and D.2, our desired result holds:

Corollary D.1. *Suppose that Theorem 1 and Assumptions 1, 2, D.1, and D.2 hold. Then*

$$\begin{aligned} \hat{\theta} - \theta &= \hat{\mathbb{E}}_N [\varphi(C_i, Y_i, X_i, \varrho_i)] + o_{\mathbb{P}}(N^{-1/2}), \\ \varphi(C_i, Y_i, X_i, \varrho_i) &:= \sum_{c=1}^C \frac{\partial h}{\partial \mu'_c} \Big|_{(\mu, \eta) = (\mu, \eta)} \psi_c(C_i, Y_i, X_i) + \frac{\partial h}{\partial \eta'} \Big|_{(\mu, \eta) = (\mu, \eta)} \varrho_i, \end{aligned} \quad (\text{D.6})$$

where φ satisfies $\mathbb{E}[\varphi(C_i, Y_i, X_i, \varrho_i)] = \mathbf{0}_p$ and $\mathbb{E}[\|\varphi(C_i, Y_i, X_i, \varrho_i)\|_2^2] < \infty$. Further, as $N \rightarrow \infty$,

$$\sqrt{N}(\hat{\theta} - \theta) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}_p, \Sigma_{\theta}), \quad \Sigma_{\theta} := \mathbb{E}[\varphi(C_i, Y_i, X_i, \varrho_i) \varphi(C_i, Y_i, X_i, \varrho_i)']. \quad (\text{D.7})$$

Corollary D.1 follows from a straightforward application of the Delta Method and a classical multivariate Central Limit Theorem.⁵⁵

Before continuing, we note that depending on the choice of h , Σ_{θ} may have zeros in its diagonal entries, in which case $\hat{\theta}$ will have zero asymptotic variance. An implication of this limiting distribution

⁵⁵See e.g. Theorem 3.1 in van der Vaart (2000).

degeneracy is that the confidence intervals $\hat{\mathcal{C}}_j$ defined in (D.5) based on the limiting Gaussian distribution (D.7) will have zero width for some coordinates and thus zero coverage of those coordinates. The following assumption rules out such knife-edge cases:

Assumption D.3. The diagonal entries of Σ_θ are all strictly positive.

To state our result on the validity of the bootstrap-based inference procedure described in Section 3 under this assumption, we define the random vector $\hat{\theta}^* = \hat{\theta}(W)$, where $\hat{\theta}(\cdot)$ is defined in (D.2), W is a random vector of weights with i th coordinate given by $W_i = \xi_i / \sum_{j=1}^N \xi_j$, and $\xi_1, \dots, \xi_N$ are draws from $\text{Exponential}(1)$ independent of both each other and $\hat{\theta}(\cdot)$. Then, under some slight generalizations of Assumptions 1 and D.1 stated in Appendix H.2 for brevity, our inference procedure satisfies the following validity guarantee:

Theorem D.2. *Suppose that Theorem 1 and Assumptions 1, 2, D.2, and D.3 hold, along with Assumptions H.1 and H.2 stated in Appendix H.2. Then the confidence intervals $\hat{\mathcal{C}}_j$ defined in (D.5) for the coordinates of θ have asymptotic simultaneous coverage at least $1 - \alpha$:*

$$\liminf_{N \rightarrow \infty} \mathbb{P} \left(\theta \in \bigtimes_{j=1}^p \hat{\mathcal{C}}_j \right) \geq 1 - \alpha.$$

We provide a proof of Theorem D.2 in Appendix H.2.

We now return to Examples D.1 and 1 and verify that Assumptions D.2 and H.2 (and therefore Assumption D.1) hold:

Example D.1 (continued). From (D.1), we can see that the relative cohort sizes p_c and cohort-specific observed outcome means m_{ct} are not included in μ and thus form the components of η we must also estimate. Of course, p_c and m_{ct} are identified and consistently estimable via simple averages of the observables $\mathbb{1}\{C_i = c\}$, $\mathbb{1}\{t \geq C_i\}Z_{it}$, and $\mathbb{1}\{t < C_i\}Y_{it}$. Thus, by standard arguments, the components of η expressed as a map from distributions to nuisance parameter values (see Appendix H.2 for details) must be Hadamard differentiable with respect to the distribution over which the expectations in their definitions are taken, implying Assumption H.2 and, by extension, the weaker Assumption D.1 hold. Since $\theta_{\text{dyn},j}$ is linear in products of the components of μ and η , Assumption D.2 immediately holds as well.

Example 1 (continued). To verify that our target parameters $\theta_{\text{col},t_1,t_2}$ and $\theta_{\text{row},t_1,t_2}$ satisfy the assumptions required for Theorem D.2 to hold, we note that, as shown in Example 1, the components of the nuisance parameter vector η simply consist of the shares of units in each cohort p_c . By standard arguments, these components of η expressed as a map from distributions to nuisance parameter values (again, see Appendix H.2 for details) are Hadamard differentiable with respect to the distributions over which the means are taken, so Assumption H.2 and the weaker Assumption D.1 hold. Further since $\theta_{\text{col},t_1,t_2}$ and $\theta_{\text{row},t_1,t_2}$ can be expressed as ratios of linear combinations of products of

the components of μ and η , it is clearly differentiable so long as

$$\mathbb{E}[Y_{it_1} \mid t_1 \in \mathcal{T}_{C_i}] - \mathbb{E}[Y_{it_1} \mid t_2 \in \mathcal{T}_{C_i}] \neq 0.$$

Thus, under this additional condition, Assumption D.2 holds as well.

E Proofs of Intermediate Results in Appendices

E.1 Proof of Theorem B.1

Consider any $\hat{M} \in \mathcal{B}(M)$. By Weyl's inequality (see e.g. Vershynin (2018, Theorem 4.5.3)),

$$\max_j |\lambda_j(M) - \lambda_j(\hat{M})| \leq \|M - \hat{M}\|_{\text{op}} \leq \Delta(M). \quad (\text{E.1})$$

Let

$$a(M) := \begin{cases} \frac{\lambda_{s+1}(M) + \lambda_s(M)}{2}, & s > 0 \\ \lambda_1(M) - 2\Delta(M), & s = 0 \end{cases}, \quad b(M) := \begin{cases} \frac{\lambda_{s+r+1}(M) + \lambda_{s+r}(M)}{2}, & s+r < d \\ \lambda_d(M) + 2\Delta(M), & s+r = d \end{cases}.$$

When $s > 0$, it must be that

$$a(M) - \lambda_s(\hat{M}) = \underbrace{\frac{\lambda_{s+1}(M) - \lambda_s(M)}{2}}_{\geq 2\Delta(M)} + \underbrace{\lambda_s(M) - \lambda_s(\hat{M})}_{\geq -\Delta(M) \text{ by (E.1)}} \geq \Delta(M),$$

and

$$\lambda_{s+1}(\hat{M}) - a(M) = \underbrace{\lambda_{s+1}(\hat{M}) - \lambda_{s+1}(M)}_{\geq -\Delta(M) \text{ by (E.1)}} + \underbrace{\frac{\lambda_{s+1}(M) - \lambda_s(M)}{2}}_{\geq 2\Delta(M)} \geq \Delta(M).$$

When $s = 0$, $a(M) - \lambda_s(\hat{M}) = \infty \geq \Delta(M)$, and

$$\lambda_{s+1}(\hat{M}) - a(M) = \underbrace{\lambda_1(\hat{M}) - \lambda_1(M)}_{\geq -\Delta(M) \text{ by (E.1)}} + 2\Delta(M) \geq \Delta(M).$$

Thus, in both cases, by the definition of $\Delta(M)$ in (B.1),

$$\min \left\{ a(M) - \max\{\lambda_s(M), \lambda_s(\hat{M})\}, \min\{\lambda_{s+1}(M), \lambda_{s+1}(\hat{M})\} - a(M) \right\} \geq \Delta(M) > 0. \quad (\text{E.2})$$

Using similar logic, we can also show that

$$\min \left\{ b(M) - \max\{\lambda_{s+r}(M), \lambda_{s+r}(\hat{M})\}, \min\{\lambda_{s+r+1}(M), \lambda_{s+r+1}(\hat{M})\} - b(M) \right\} \geq \Delta(M) > 0. \quad (\text{E.3})$$

By (E.43) and (E.44), we also prove bounds on the operator norms of $\mathcal{A}(\Pi_0)$ and $\mathcal{B}(\Pi_0)$ that are uniform in Π_0 .

F Supplementary Discussions and Results

F.1 Oracle O³ Algorithm Convergence Implies Full-Rank Cohort-Specific Factor Matrices

If the oracle O³ Algorithm discussed in Footnote 29 converges to a single super cohort, i.e. $M^{(K)} = 1$, there must exist some iteration K_c of the oracle O³ Algorithm where there exists a super cohort m_1 such that $\mathcal{T}_c = \mathcal{T}_{\mathcal{S}_{m_1}^{(K_c)}}$ and such that super cohort m_1 is connected to some other super cohort m_2 in the O³ Graph $\mathcal{G}_r^{(K_c+1)}$ constructed in the subsequent iteration. Since $E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(K_c)}}}$ is a square matrix, the column space of $\Gamma' E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(K_c)}}}$ must be a subset of the column space of $\Gamma' E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(K_c)}}} E_{\mathcal{T}_{\mathcal{S}_{m_2}^{(K_c)}}}$, which, along with (3), implies that

$$\begin{aligned} \text{rank}(\underbrace{E_c \Gamma}_{T \times r}) &= \text{rank}(\Gamma' E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(K_c)}}}) && (T > r, \text{rank is } \min\{\text{row rank}, \text{col rank}\}, \mathcal{T}_c = \mathcal{T}_{\mathcal{S}_{m_1}^{(K_c)}}) \\ &\geq \text{rank}(\Gamma' E_{\mathcal{T}_{\mathcal{S}_{m_1}^{(K_c)}}} E_{\mathcal{T}_{\mathcal{S}_{m_2}^{(K_c)}}}) \\ &= r && (m_1 \text{ and } m_2 \text{ are neighbors in } \mathcal{G}_r^{(K_c+1)}, \text{ so (3) holds}). \end{aligned}$$

Since $E_c \Gamma$ is $T \times r$ with $T > r$, $\text{rank}(E_c \Gamma) \leq r$ must also hold.

F.2 Equivalence of Graph-Based Identification Criteria in Panel Data Models and O³ Algorithm Convergence to a Single Super Cohort

Several papers in the literature on bipartite match data develop approaches for identifying μ_{ct} (or functions of it) under other models of how unobserved confounders affect outcomes Y_{it}^* that still impose strict exogeneity and fixed-effect-like assumptions (Bonhomme, 2020). To prove identification of μ_{ct} under their alternative models, these papers appeal to the connectedness of different graphs with nodes that represent units and/or outcomes in the nomenclature of this paper. In this section, we show that two types of graphs and accompanying assumptions about their connectedness made in this literature are both equivalent to the O³ Algorithm (Algorithm 2) converging to a single super cohort, i.e. $M^{(K)} = 1$, when $r = 1$.

First, we consider the identification arguments in Abowd et al. (2002) and Jochmans and Weidner (2019), which study TWFE models of outcomes $\mathbb{E}[Y_{it}^* \mid \lambda_i, C_i] = \gamma_t + \lambda_i$, where γ_t and λ_i are both one-dimensional. These papers condition on a sample of units and the set of their uni-dimensional fixed effects $\{\lambda_i\}_{i=1}^N$, and they show that under appropriate normalizations of the fixed effects, the TWFE regression estimators $\hat{\gamma}_t$ of the outcome fixed effects γ_t converge to an appropriately normal-

ized instance of the corresponding outcome fixed effects $\tilde{\gamma}_t$ at a $N^{-1/2}$ rate as $N \rightarrow \infty$ under two assumptions. First, they assume that a finite-population variant of Assumption 2 holds so that a non-vanishing fraction of units have outcome t observed. Second, they define the bipartite graph $\tilde{G}$ consisting of nodes corresponding to units and outcomes and an edge between any unit i and outcome t for which outcome Y_{it}^* is observed and assume that $\tilde{G}$ is connected.

As it turns out, this assumption is equivalent to the O^3 Algorithm converging to a single super cohort when $r = 1$:

Proposition F.1. *When $r = 1$ and every entry of Γ is non-zero, the bipartite graph $\tilde{G}$ being connected implies that the O^3 Algorithm converges to a single super cohort after a single iteration, i.e. $K = 1$ and $M^{(1)} = 1$. If, in addition, Assumption 2 holds, then the O^3 Algorithm converging to a single super cohort after a single iteration implies that $\tilde{G}$ is connected with high probability.*

We provide a proof of Proposition F.1 in Appendix H.3. We note that since when $r = 1$, Γ is a vector, the assumption that every entry of Γ is non-zero is just assuming that the systematic component $\gamma_t \lambda_i$ in the factor model (1) matters for determining every potential outcome.

Second, we consider the identification arguments in Bonhomme et al. (2019), which identifies the distributions of Y_{it} under a model of outcomes with discrete unobserved heterogeneity, and Hull (2018) and Section 3.3 of Kline (2024), which identify differences in μ_{ct} across t under a TWFE-like model of outcomes. Unlike Abowd et al. (2002) and Jochmans and Weidner (2019) and similarly to this paper, these papers model an infinite population of units and, due to their focus on bipartite match outcomes, take seriously the fact that units form one match at a time sequentially. Abstracting away from the time dimension to match the setup in this paper, at their core, Bonhomme et al. (2019), Hull (2018), and Kline (2024) base their identification arguments off of variants of the assumption that another graph $\check{G}$ is connected, where $\check{G}$ is an undirected graph whose nodes correspond to outcomes and whose edges between two outcomes t_1 and t_2 exist if a positive measure of units have both outcomes t_1 and t_2 observed.

Similarly to Proposition F.1, we can also show that the assumption that $\check{G}$ is connected is equivalent to the O^3 Algorithm converging to a single super cohort when $r = 1$:

Proposition F.2. *When $r = 1$ and every entry of Γ is non-zero, the graph $\check{G}$ being connected implies that the O^3 Algorithm converges to a single super cohort after a single iteration, i.e. $K = 1$ and $M^{(1)} = 1$. If, in addition, Assumption 2 holds, then the O^3 Algorithm converging to a single super cohort after a single iteration implies that $\check{G}$ is connected with high probability.*

We provide a proof of Proposition F.2 in Appendix H.4.

F.3 Asymptotic Linearity of the Principal Components Estimator

As discussed intuitively in Section 4.3, the Principal Components (PC) estimator of cohort-specific factors can be shown to identify the column space of the cohort-specific factor matrix $E_c \Gamma$. In this

section, we formalize this fact under slightly weaker assumptions than those typically imposed in the literature. To define the PC estimator formally, let V_c denote the population covariance matrix of the observed outcomes $E_c Y_i$ for cohort c :

$$V_c := \mathbb{E}[(E_c Y_i - E_c \mu_c)(E_c Y_i - E_c \mu_c)' \mid C_i = c], \quad (\text{F.1})$$

and let $\hat{V}_c$ denote its empirical counterpart:

$$\hat{V}_c := \hat{\mathbb{E}}_N[(E_c Y_i - \hat{\mathbb{E}}_N[E_c Y_i \mid C_i = c])(E_c Y_i - \hat{\mathbb{E}}_N[E_c Y_i \mid C_i = c])' \mid C_i = c];$$

the PC estimator $\Pi(\hat{\Gamma}_{c,PC})$ of $\Pi(E_c \Gamma)$ is defined as the projection matrix onto the span of the eigenvectors of $\hat{V}_c$ corresponding to $\hat{V}_c$'s r largest eigenvalues.

Under two assumptions that we define next, we will show that a population analog of the PC estimator, i.e. the projection matrix onto the span of the eigenvectors of V_c corresponding to V_c 's r largest eigenvalues, identifies $\Pi(E_c \Gamma)$, as required for Theorem 1 to hold. The first assumption guarantees that all of the loadings' dimensions are non-trivial:

Assumption F.1. For every cohort $c \in [C]$, the loading covariance matrix $\text{Cov}(\lambda_i \mid C_i = c)$ has finite operator norm and is positive definite, $\mathbb{E}[\|\lambda_i\|_2^4 \mid C_i = c] < \infty$ and $\mathbb{E}[\|\varepsilon_i\|_2^4 \mid C_i = c] < \infty$.

Our other assumption is that the outcome residuals are uncorrelated across outcomes and have the same variance within each unit:

Assumption F.2. There exist positive random variables σ_i^2 defined on the same probability space as (C_i, Y_i^*) such that $\mathbb{E}[\varepsilon_i \varepsilon_i' \mid \sigma_i^2, C_i] = \sigma_i^2 I$ almost surely and $\sigma_c^2 := \mathbb{E}[\sigma_i^2 \mid C_i = c] < \infty$ for every cohort $c \in [C]$.

Before continuing, we provide one more definition. Let

$$s_{1c}^2 \leq \dots \leq s_{rc}^2$$

denote the smallest through largest eigenvalues of the matrix $\Gamma' E_c \Gamma \text{Cov}(\lambda_i \mid C_i = c)$. We note that $s_{1c}^2 > 0$ by Assumption F.1 and the fact that $E_c \Gamma$ is full-rank when the factor vectors are in general position. We are now equipped to state our identification result:

Lemma F.3. Suppose the factor vectors $\{\gamma_t : t \in [T]\}$ are in general position, Assumptions F.1 and F.2 hold, and let $\tilde{\Gamma}_{c,PC}$ be any $T \times r$ matrix whose columns are eigenvectors corresponding to the r largest eigenvalues of V_c , defined in (F.1). Then $\Pi(\tilde{\Gamma}_{c,PC}) = \Pi(E_c \Gamma)$, meaning $\Pi(E_c \Gamma)$ is identified as required for Theorem 1 to hold. Further, V_c 's eigenvalues ordered from smallest to largest are $T - T_c$ zeros followed by $T_c - r$ repetitions of σ_c^2 followed by $s_{c1}^2 + \sigma_c^2, \dots, s_{cr}^2 + \sigma_c^2$.

We provide a proof of Lemma F.3 in Appendix H.5.

Next, we show that $\Pi(\hat{\Gamma}_{c,PC})$ is a consistent and asymptotically linear estimator of $\Pi(E_c \Gamma)$. To introduce this result, we let $\tilde{\gamma}_{cj}$ denote an eigenvector of V_c corresponding to the j th smallest eigenvalue of V_c .

Proposition F.4. Suppose the factor vectors $\{\gamma_t : t \in [T]\}$ are in general position and Assumptions 2, F.1, and F.2, hold. Then

$$\left\| [\Pi(\hat{\Gamma}_{c,PC}) - \Pi(E_c\Gamma)] - \hat{\mathbb{E}}_N [\phi_{c,PC}(C_i, Y_i)] \right\|_{\text{op}} = \mathcal{O}_{\mathbb{P}} \left(s_{1c}^{-2} [s_{1c}^{-2} + r(T_c - r)] N^{-1} p_c^{-1} \right), \quad (\text{F.2})$$

where

$$\phi_{c,PC}(C_i, Y_i) := \sum_{j=T-r+1}^T \sum_{k=T-T_c+1}^{T-r} \frac{\mathbb{1}\{C_i = c\}}{p_c \cdot s_{(j-(T-r))c}^2} \cdot g(\Pi(\tilde{\gamma}_{cj}), (V_c - E_c(Y_i - \mu_c)(Y_i - \mu_c)'E_c), \Pi(\tilde{\gamma}_{ck})),$$

and $\phi_{c,PC}$ satisfies $\mathbb{E}[\phi_{c,PC}(C_i, Y_i)] = \mathbf{0}_{T^2}$ and

$$\mathbb{E}[\|\phi_{c,PC}(C_i, Y_i)\|_{\text{op}}^2] \leq \frac{64r(T_c - r)}{p_c s_{1c}^2} \|\Gamma\|_{\text{op}}^4 (\mathbb{E}[\|\lambda_i\|_2^4 \mid C_i = c] + \mathbb{E}[\|\varepsilon_i\|_2^4 \mid C_i = c]) < \infty.$$

Thus, Assumption 1 holds.

We provide a proof of Proposition F.4 in Appendix H.6. We note that as a consequence of (H.23), Assumption H.1 holds as well, although for brevity, we do not introduce the additional notation necessary to state such a result formally here.

We conclude this section with a more detailed discussion of why the assumptions needed for Proposition F.4 to hold are slightly weaker than existing consistency and asymptotic linearity results for the PC estimator in the literature of which we are aware. It has been shown that the PC estimator is consistent and asymptotic linear when T remains finite as N grows under Assumptions F.1 and F.2 in the sense that

$$\hat{\Gamma}_{c,PC} - E_c\Gamma\hat{Q} = \hat{\mathbb{E}}_N [\tilde{\phi}_{c,PC}(C_i, Y_i)] + o_{\mathbb{P}}(N^{-1/2}) \quad (\text{F.3})$$

for some mean-zero influence function $\tilde{\phi}_{c,PC}$ and random matrix $\hat{Q}$ such that $\hat{Q} \xrightarrow{\mathbb{P}} Q$ for some deterministic matrix Q (see e.g. Theorem 5 in J. Bai (2003)).⁵⁶ Such results require the eigenvalues $s_{1c}^2, \dots, s_{rc}^2$ to be distinct.⁵⁷

However, our theory in Section 4.2 only requires assumptions that guarantee the consistency of $\Pi(\hat{\Gamma}_{c,PC})$ as an estimator of $\Pi(E_c\Gamma)$, which in turn only requires the existence of a random basis matrix $\hat{Q}$ such that $\hat{\Gamma}_{c,PC}\hat{Q}' = E_c\Gamma + \mathcal{O}_{\mathbb{P}}(N^{-1/2})$, not that $\hat{Q}$ has a deterministic probability limit, as in (F.3).⁵⁸ As such, we do not require distinctness of the eigenvalues $s_{1c}^2, \dots, s_{rc}^2$ to show that the population equivalent of $\hat{\Gamma}_{c,PC}$ identifies $\Pi(E_c\Gamma)$ as required for Theorem 1 to hold and that $\hat{\Gamma}_{c,PC}$

⁵⁶Theorem 1 in J. Bai and Ng (2002) does show a similar result to the statement $\sqrt{N}\|\hat{\Gamma}_{c,PC} - E_c\Gamma\hat{Q}\|_F^2 = \mathcal{O}_{\mathbb{P}}(1)$ without requiring $\hat{Q}$ to have a probability limit, but they prove it in the more general case where ε_{it} is allowed to be heteroskedastic and weakly dependent, so in their theorem statement the $\sqrt{N}$ factor is replaced with $\min\{\sqrt{N}, \sqrt{T}\}$ and they require $\min\{N, T\} \rightarrow \infty$.

⁵⁷See Assumption G in J. Bai (2003) and Assumption A2(iii) in J. Bai and Ng (2023) for examples of such eigenvalue uniqueness conditions. After introducing Assumption G, J. Bai (2003) notes that such an assumption is not necessary to show that consistent estimators exist for identifiable quantities derived from the factor model.

⁵⁸To see why, note that if such a $\hat{Q}$ did not exist, then for all potentially random basis matrices $\hat{Q}$, $\Pi(\hat{\Gamma}_{c,PC}) = \Pi(\hat{\Gamma}_{c,PC}\hat{Q}') \neq \Pi(E_c\Gamma) + \mathcal{O}_{\mathbb{P}}(N^{-1/2})$, which would be a violation of Proposition F.4.

itself satisfies Assumptions 1 and H.1 required for our estimator to be consistent and to be useful for valid bootstrap inference.

G Empirical Illustration: More Details

G.1 Dataset Construction Details

As discussed in Section 5.1, to account for within-province firm heterogeneity in a flexible way, we cluster the firms located in each province into $k = 3$ types using the k -means-based procedure proposed in Bonhomme et al. (2019). In particular, we let F be the number of firms in our sample, we let N_f denote the number of workers who ever worked for firm f in 1998 and 1999, we let $\mathcal{F}_p \subset \{1, \dots, F\}$ denote the subset of firms located in province p , and we let $\hat{G}_{Y_f}$ denote the empirical cumulative distribution function of average weekly wages in 1998 and 1999 across workers that ever worked for firm $f \in \{1, \dots, F\}$ during 1998 and 1999. To cluster the firms $\mathcal{F}_p$ within province p , we solve the following clustering problem for each province p , as in Bonhomme et al. (2019):

$$\min_{\{k(f) : f \in \mathcal{F}_p\}, H_{p1}, \dots, H_{pK}} \sum_{f \in \mathcal{F}_p} N_f \int \left(\hat{G}_{Y_f}(y) - H_{pk(f)}(y) \right)^2 d\mu(y), \quad (\text{G.1})$$

where μ is a discrete uniform measure with mass points at 100 evenly spaced quantiles of the distribution of average weekly wages across all firms in the sample. For each province p , we solve (G.1) 10 times with different initial cluster assignments and select the clustering solution with the lowest objective value. We plot the number of firms in each of the final clusters within each province in Figure G.1 and the distribution of cohort sizes under this clustering in Figure G.2. We also report summary statistics for the final panel constructed from the VWH dataset based on the sequence of sample restrictions we make to construct the final panel described in Section 5.1 in Table G.1.